Project: Worked examples
Title: O-ring and seal gland
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1. Scope and method

Calculates squeeze, stretch, gland fill and extrusion gap of a piston, rod or face O-ring seal at the cold and hot ends of its temperature and swell range. You supply the ring sizes, bands and extrusion limits.

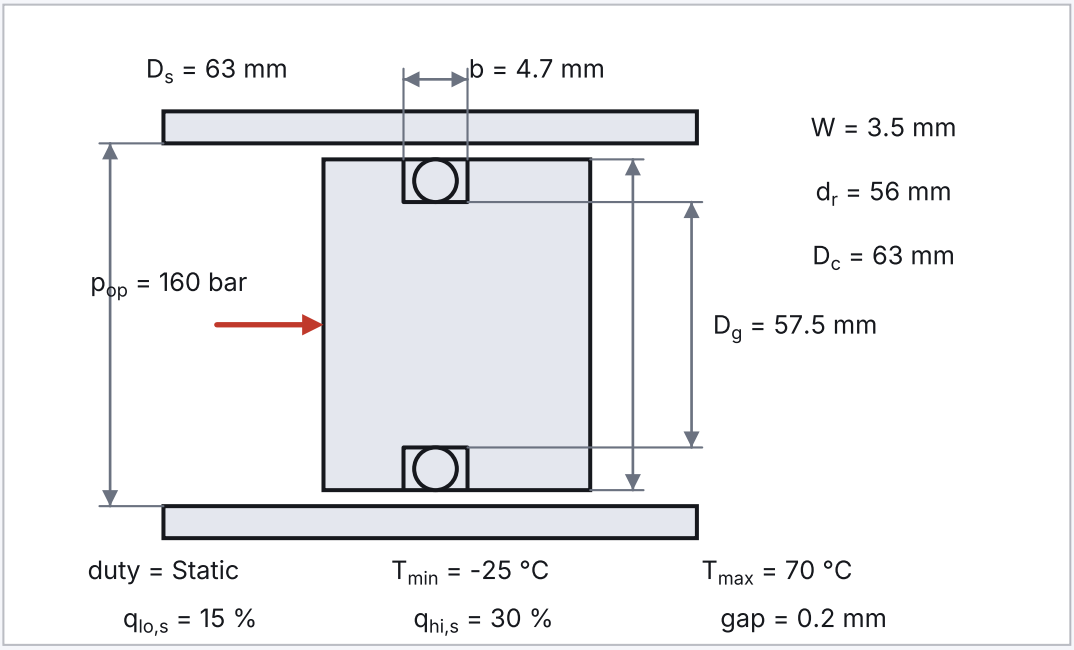


Figure 1 — A piston seal in section: the ring of section W in the piston groove (root D_g, width b), the piston D_c in the bore D_s, pressure p_{op} on one side. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Ring stretch	$X = (g \cdot D_{ref})/(f \cdot d) - 1$, D_{ref} the diameter the ring ID sits on (groove root for a piston or face seal, rod for a rod seal); $f = (1 + \alpha(T - 20\ °C))(1 + swell)^{(1/3)}$ the ring and $g = 1 + \alpha_m(T - 20\ °C)$ the groove metal
Ring cross-section	$W_{eff} = f \cdot W \cdot (1 - r(X))/\sqrt{(1 - c)}$, r(X) the chosen relation (none; $1 - 1/\sqrt{(1+X)}$; $Y = 0.56 + 0.59X - 0.0046X^2$ per cent as cited, used for about 3 to 25 % stretch only; $v \cdot X$) and c the compression of the ring outside diameter where it is held (rod seal in the housing groove; internal-pressure face seal on the groove outer wall), $c = 1 - g \cdot D_g/(f \cdot (d + 2W))$
Depth and gap	piston $h = (D_s - D_g)/2$, $gap = D_s - D_c$; rod $h = (D_g - D_s)/2$, $gap = D_c - D_s$; face h and gap entered directly (all times g)
Ring squeeze	$q = 1 - g \cdot h/W_{eff}$, minimum at the cold end from the largest depth over the smallest W_{eff}, maximum at the hot end

WHAT IT CALCULATES	RELATION
Gland fill	ring area over the groove area $g \cdot b \cdot g \cdot h - 2(1 - \pi/4)(g \cdot r_g)^2$; maximum at the hot end on a volume basis, $(\pi/4) \cdot (f \cdot W)^2 \cdot (f/g)$ with $f^3 = (1 + \alpha \Delta T)^3 (1 + s)$ (an outside diameter held in the groove: $(\pi/4) \cdot (f \cdot W)^2 / (1 - c)$), minimum at the cold end $(\pi/4) \cdot W_{eff}^2$
Worst case	each range from the engine's worst-case stack of the dimension ($\Sigma \lvert coefficient \rvert \cdot$half-band), combined at the corner that is worst for the quantity
Monte Carlo	the engine's seeded Monte Carlo on the exact criterion, each dimension from its own distribution; the fraction below the lower and above the upper limit. The capability index C_{pk} uses the criterion linearised about the mean dimensions (secant slope across each band)
Extrusion gap	the largest gap over the installed, cold and hot temperatures (the sealing-surface part and the groove metal each with its own expansion) plus the bore dilation under pressure, against the permitted gap from your record (interpolated in hardness and pressure) or the value entered; with neither the check fails (not verified)

Basis: Cross-section reduction with stretch (relation quoted second-hand; not confirmed)[1](§3.5); O-ring installation guidance (stretch rule quoted second-hand; not confirmed)[2]; O-rings: inside diameters, cross-sections, tolerances and size codes[3]; O-rings: housing dimensions for general applications[4].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Gland arrangement	`Piston (radial)`		spec: Cylinder datasheet LC-63: ring on the piston, sealing on the bore	any
duty	Static or dynamic duty (selects the squeeze band)	`Static`		spec: Cylinder datasheet LC-63: locking duty, no stroke under pressure	any
$face_p$	Face seal only: which side the pressure acts on (external: the ring ID seats on the groove inner wall and is stretched; internal: the ring OD seats on the groove outer wall and is compressed)	`External (ring ID seated on the groove)`		assumption: Face seals only; this piston seal does not use it, so the template default is kept.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d_r	Ring inside diameter d (mean of the supplier's tolerance)	56	mm	reference: record: oring.ring, illustrative ring size invented for this example, to be replaced by the supplier's catalogue size	3 … 500 mm
$d_{r,tol}$	Ring inside diameter tolerance, plus or minus	0.45	mm	reference: record: oring.ring, illustrative tolerance invented for this example	0 … 2 mm
W	Ring cross-section diameter	3.5	mm	reference: record: oring.ring, illustrative cross-section invented for this example	1 … 10 mm
W_{tol}	Ring cross-section tolerance, plus or minus	0.1	mm	reference: record: oring.ring, illustrative tolerance invented for this example	0 … 0.4 mm
D_s	Sealing-surface diameter: the bore of a piston seal, the rod of a rod seal (not used for a face seal)	63	mm	drawing: LC-63-001 rev B	5 … 1000 mm
$D_{s,u}$	Sealing-surface diameter, upper deviation	0.046	mm	drawing: LC-63-001 rev B	-0.5 … 0.5 mm
$D_{s,l}$	Sealing-surface diameter, lower deviation	0	mm	drawing: LC-63-001 rev B	-0.5 … 0.5 mm
D_g	Groove root diameter: on the piston, in the housing of a rod seal, or for a face seal the diameter the ring seats on (the groove inner wall for external pressure, the groove outer wall for internal pressure)	57.5	mm	drawing: LC-63-002 rev B	5 … 1000 mm
$D_{g,u}$	Groove root diameter, upper deviation	0	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
$D_{g,l}$	Groove root diameter, lower deviation	-0.074	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
D_c	Mating part that sets the extrusion gap: the piston outside diameter, or the housing bore round the rod (not used for a face seal)	63	mm	drawing: LC-63-002 rev B	5 … 1000 mm
$D_{c,u}$	Mating part diameter, upper deviation	-0.03	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
$D_{c,l}$	Mating part diameter, lower deviation	-0.06	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
h_f	Face-seal groove depth (face seal only), where the ring is compressed over the depth plus the gap	2.7	mm	assumption: Face seals only; this piston seal does not use it, so the template default is kept.	0.3 … 20 mm
$h_{f,u}$	Face-seal groove depth, upper deviation	0.05	mm	assumption: Face seals only; this piston seal does not use it, so the template default is kept.	-0.5 … 0.5 mm
$h_{f,l}$	Face-seal groove depth, lower deviation	0	mm	assumption: Face seals only; this piston seal does not use it, so the template default is kept.	-0.5 … 0.5 mm
e_f	Face-seal gap between the flanges (face seal only), which is the extrusion gap and adds to the compression height	0.05	mm	assumption: Face seals only; this piston seal does not use it, so the template default is kept.	0 … 2 mm
$e_{f,u}$	Face-seal gap, upper deviation	0.05	mm	assumption: Face seals only; this piston seal does not use it, so the template default is kept.	-0.5 … 0.5 mm
$e_{f,l}$	Face-seal gap, lower deviation	0	mm	assumption: Face seals only; this piston seal does not use it, so the template default is kept.	-0.5 … 0.5 mm
b	Groove width	4.7	mm	drawing: LC-63-002 rev B	1 … 20 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
b_u	Groove width, upper deviation	0.1	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
b_l	Groove width, lower deviation	0	mm	drawing: LC-63-002 rev B	-0.5 … 0.5 mm
ecc	Gland eccentricity or run-out (one-sided offset; it deepens the gland by this much where the squeeze is least and shallows it where the squeeze is most; 0 if none)	0	mm	assumption: Gland taken as concentric (no run-out) for this example; enter the run-out from the drawing when it is known.	0 … 0.5 mm
r_g	Groove bottom corner radius	0.25	mm	drawing: LC-63-002 rev B	0 … 1 mm
T_{min}	Lowest operating temperature (the ring sits at 20 °C when installed)	-25	°C	spec: Hydraulic specification HS-63: oil temperature range −25 to +70 °C	-60 … 100 °C
T_{max}	Highest operating temperature	70	°C	spec: Hydraulic specification HS-63: oil temperature range −25 to +70 °C	-40 … 200 °C
α	Ring linear thermal expansion coefficient α (an indicative elastomer figure; take yours from the compound or the material library)	0.00015	1/K	assumption: Indicative elastomer figure; replace by the compound's value or a material from the library.	0.00005 … 0.0003 1/K
α_m	Groove metal linear thermal expansion coefficient α_m (the part that carries the groove; one material record per sheet, so this one is entered from the metal's data sheet)	0.000012	1/K	assumption: Steel groove metal, indicative.	0.000005 … 0.00003 1/K
s_{min}	Smallest volume change of the ring in the fluid (negative for shrinkage; 0 if none)	0	%	assumption: No shrinkage assumed in the mineral oil.	-10 … 0 %

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
s_{max}	Largest volume swell of the ring in the fluid	4	%	assumption: Volume swell of the compound in the oil, assumed 4 %; replace by the compound's immersion data.	0 … 25 %
red	Cross-section reduction with stretch	Incompressible rubber (derived)		assumption: Constant-volume rubber relation, derived.	any
v	Poisson ratio of the ring, for the linear relation	0.49		assumption: Rubber taken as nearly incompressible, v = 0.49; take it from the compound data.	0.3 … 0.5
$q_{lo,s}$	Static duty: lowest acceptable squeeze (from your handbook)	15	%	reference: record: oring.criteria, static squeeze band, to be taken from the owner's handbook or the seal maker	5 … 30 %
$q_{hi,s}$	Static duty: highest acceptable squeeze	30	%	reference: record: oring.criteria, static squeeze band, to be taken from the owner's handbook or the seal maker	10 … 45 %
$q_{lo,d}$	Dynamic duty: lowest acceptable squeeze	10	%	assumption: Dynamic duty only; this static seal does not use it, so the template default is kept.	5 … 25 %
$q_{hi,d}$	Dynamic duty: highest acceptable squeeze	25	%	assumption: Dynamic duty only; this static seal does not use it, so the template default is kept.	10 … 35 %
$fill_{lo}$	Gland fill: lowest acceptable	60	%	reference: record: oring.criteria, gland fill band (default 60 to 85 %)	40 … 80 %
$fill_{hi}$	Gland fill: highest acceptable	85	%	reference: record: oring.criteria, gland fill band (default 60 to 85 %)	60 … 95 %

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
x_{lo}	Installed stretch: lowest acceptable (negative allows a ring that is compressed on its seat)	0	%	reference: record: oring.criteria, installed stretch lower limit	-5 … 5 %
x_{hi}	Installed stretch: highest acceptable (the installation rule is up to 6 % on a piston groove); also applied at the cold end	6	%	reference: record: oring.criteria, installed stretch limit (up to 6 % on a piston groove, as quoted second-hand)	1 … 10 %
od_{hi}	Ring outside-diameter compression: highest acceptable, in a rod-seal groove or against the groove outer wall of an internal-pressure face seal (your record; the default is a placeholder equal to the stretch limit, not a handbook value)	6	%	assumption: Rod seals and internal-pressure face seals only; this piston seal does not use it, so the template default is kept.	1 … 10 %
H_A	Ring hardness, Shore A (for the extrusion record)	70		spec: Seal specification: compound 70 Shore A	40 … 95
p_{op}	Operating pressure (for the extrusion record, gauge)	160	bar g	spec: Hydraulic specification HS-63: working pressure 160 bar	0 … 700 bar g
dD_p	Diametral growth under the operating pressure of the bore (piston seal) or of the housing bore round the rod (rod seal), from your cylinder or housing calculation; added to the extrusion gap (0 if not known)	0	mm	assumption: Bore growth under pressure not known for this example, so none is added to the gap; take it from the cylinder calculation.	0 … 0.5 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
gap_{allow}	Largest permitted extrusion gap at this pressure and hardness, if you have it as one value: diametral (the clearance $D_s - D_c$) for a radial seal, axial for a face seal; a radial chart value is half of it	0.2	mm	reference: record: oring.extrusion, illustrative permitted gap invented for this example, to be replaced by the seal maker's value for 70 Shore A at 160 bar	0.01 ... 1 mm
n_{mc}	Monte Carlo samples	20000		assumption: 20 000 samples.	1000 ... 100000
seed	Monte Carlo seed	42		assumption: Fixed seed for a repeatable sheet.	0 ... 1000000
k_{sigma}	Standard deviations per half-band (normal tolerances and measured normals)	3		assumption: Normal tolerances taken as ±3σ.	1 ... 6
dist	Distribution of the toleranced dimensions	Normal		assumption: Normal distribution of the machined dimensions.	any

3. Assumptions

A1. The gland is a plain rectangular groove with a bottom corner radius. (Closed-form gland geometry; no gland table.)
A2. The ring is a circle of the stated cross-section. (Closed-form gland geometry; no gland table.)
A3. Edge breaks, groove wall angles and surface finish are not modelled. (Closed-form gland geometry; no gland table.)
A4. Squeeze is the compressed fraction of the effective cross-section W_{eff}. (The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A5. W_{eff} is the nominal section after thermal expansion, swell and the cross-section reduction of the stretch. (The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A6. The reduction relation is your choice, and the default is the constant-volume rubber relation. (The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A7. The quadratic from the Parker handbook is implemented as quoted second-hand and is not confirmed against the handbook. (The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A8. The quadratic is used for about 3 to 25 % stretch only. (The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A9. Below 3 % the reduction is blended linearly to zero. (The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A10. Above 25 % the reduction is held at its 25 % value. (The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A11. Where the ring outside diameter is held, the compression c thickens the section to W/√(1 − c). (The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A12. A rod seal in its housing groove and an internal-pressure face seal hold the ring outside diameter.

(The relation is an input choice, and the default is the constant-volume rubber relation, derived.)
A13. The cold end is the lowest temperature with the least swell, and it sets the minimum squeeze and fill. (Bounding cases of a monotonic model.)
A14. The hot end is the highest temperature with the most swell, and it sets the maximum squeeze and fill. (Bounding cases of a monotonic model.)
A15. This assumes the ring expands more than the groove metal (α above α_m). (Bounding cases of a monotonic model.)
A16. The worst case combines each dimension at its extreme independently. (Independent ends of linear stacks.)
A17. Where one dimension appears twice, the worst case is slightly conservative. (Independent ends of linear stacks.)
A18. The groove root diameter sets both the depth and the stretch of a piston seal. (Independent ends of linear stacks.)
A19. The Monte Carlo samples each dimension from its own distribution about its mean. (Exact model on the engine's seeded sampler.)
A20. It evaluates each criterion exactly, with the non-linear model, on the seeded sampler of the engine. (Exact model on the engine's seeded sampler.)
A21. It counts the fraction outside each limit at the cold or hot end. (Exact model on the engine's seeded sampler.)
A22. It does not model pressure, friction or compression set. (Exact model on the engine's seeded sampler.)
A23. The capability index C_{pk} uses the first-order (secant) linearisation. (Exact model on the engine's seeded sampler.)
A24. The gland fill at the hot end is on a volume basis. (Volume conservation of the held ring.)
A25. A ring held circumferentially by the groove takes the whole volume change of thermal expansion and swell in its section. (Volume conservation of the held ring.)
A26. That change is $(1 + \alpha\Delta T)^3(1 + s)$, while the groove circumference grows with the groove metal. (Volume conservation of the held ring.)
A27. This is higher than scaling the section area isotropically. (Volume conservation of the held ring.)
A28. The extrusion gap is the largest diametral clearance of a radial seal over the installed, cold and hot temperatures. (Extrusion limits are licensed data and are not shipped.)
A29. The extrusion gap of a face seal is its axial gap. (Extrusion limits are licensed data and are not shipped.)
A30. The sealing-surface part and the groove metal each have their own expansion coefficient. (Extrusion limits are licensed data and are not shipped.)
A31. The gap adds the diametral growth of the bore under pressure that you enter, or 0 if none. (Extrusion limits are licensed data and are not shipped.)
A32. The permitted gap is your record, and the sheet never states a handbook value. (Extrusion limits are licensed data and are not shipped.)
A33. With no permitted gap entered, the extrusion check fails (not verified) while the operating pressure is above zero. (Extrusion limits are licensed data and are not shipped.)
A34. $face_p$ (Face seal only: which side the pressure acts on (external: the ring ID seats on the groove inner wall and is stretched; internal: the ring OD seats on the groove outer wall and is compressed)) is taken as External (ring ID seated on the groove). (Face seals only; this piston seal does not use it, so the template default is kept.)
A35. h_f (Face-seal groove depth (face seal only), where the ring is compressed over the depth plus the gap) is taken as 2.7 mm. (Face seals only; this piston seal does not use it, so the template default is kept.)
A36. $h_{f,u}$ (Face-seal groove depth, upper deviation) is taken as 0.05 mm. (Face seals only; this piston seal does not use it, so the template default is kept.)
A37. $h_{f,l}$ (Face-seal groove depth, lower deviation) is taken as 0 mm. (Face seals only; this piston seal does not use it, so the template default is kept.)
A38. e_f (Face-seal gap between the flanges (face seal only), which is the extrusion gap and adds to the

compression height) is taken as 0.05 mm. (Face seals only; this piston seal does not use it, so the template default is kept.)
A39. $e_{f,u}$ (Face-seal gap, upper deviation) is taken as 0.05 mm. (Face seals only; this piston seal does not use it, so the template default is kept.)
A40. $e_{f,l}$ (Face-seal gap, lower deviation) is taken as 0 mm. (Face seals only; this piston seal does not use it, so the template default is kept.)
A41. ecc (Gland eccentricity or run-out (one-sided offset; it deepens the gland by this much where the squeeze is least and shallows it where the squeeze is most; 0 if none)) is taken as 0 mm. (Gland taken as concentric (no run-out) for this example; enter the run-out from the drawing when it is known.)
A42. α (Ring linear thermal expansion coefficient α (an indicative elastomer figure; take yours from the compound or the material library)) is taken as 0.00015 1/K. (Indicative elastomer figure; replace by the compound's value or a material from the library.)
A43. α_m (Groove metal linear thermal expansion coefficient α_m (the part that carries the groove; one material record per sheet, so this one is entered from the metal's data sheet)) is taken as 0.000012 1/K. (Steel groove metal, indicative.)
A44. s_{min} (Smallest volume change of the ring in the fluid (negative for shrinkage; 0 if none)) is taken as 0 %. (No shrinkage assumed in the mineral oil.)
A45. s_{max} (Largest volume swell of the ring in the fluid) is taken as 4 %. (Volume swell of the compound in the oil, assumed 4 %; replace by the compound's immersion data.)
A46. red (Cross-section reduction with stretch) is taken as Incompressible rubber (derived). (Constant-volume rubber relation, derived.)
A47. ν (Poisson ratio of the ring, for the linear relation) is taken as 0.49. (Rubber taken as nearly incompressible, ν = 0.49; take it from the compound data.)
A48. $q_{lo,d}$ (Dynamic duty: lowest acceptable squeeze) is taken as 10 %. (Dynamic duty only; this static seal does not use it, so the template default is kept.)
A49. $q_{hi,d}$ (Dynamic duty: highest acceptable squeeze) is taken as 25 %. (Dynamic duty only; this static seal does not use it, so the template default is kept.)
A50. od_{hi} (Ring outside-diameter compression: highest acceptable, in a rod-seal groove or against the groove outer wall of an internal-pressure face seal (your record; the default is a placeholder equal to the stretch limit, not a handbook value)) is taken as 6 %. (Rod seals and internal-pressure face seals only; this piston seal does not use it, so the template default is kept.)
A51. dD_p (Diametral growth under the operating pressure of the bore (piston seal) or of the housing bore round the rod (rod seal), from your cylinder or housing calculation; added to the extrusion gap (0 if not known)) is taken as 0 mm. (Bore growth under pressure not known for this example, so none is added to the gap; take it from the cylinder calculation.)
A52. n_{mc} (Monte Carlo samples) is taken as 20000. (20 000 samples.)
A53. seed (Monte Carlo seed) is taken as 42. (Fixed seed for a repeatable sheet.)
A54. k_{sigma} (Standard deviations per half-band (normal tolerances and measured normals)) is taken as 3. (Normal tolerances taken as ±3σ.)
A55. dist (Distribution of the toleranced dimensions) is taken as Normal. (Normal distribution of the machined dimensions.)
A56. Static duty: the squeeze band for static seals from the oring.criteria record is used. (The bands are the user's record.)

4. Working

[42] v_W = mid ± half-band, σ (normal)
= 3.5 mm nominal → mid 3.5 ± 0.1 mm, σ 0.033333 mm
= 3.5 mm
ring cross-section W
[43] v_d = mid ± half-band, σ (normal)
= 56 mm nominal → mid 56 ± 0.45 mm, σ 0.15 mm
= 56 mm

ring inside diameter d

[44] v_{Ds} = mid ± half-band, σ (normal)
= 63 mm nominal → mid 63.023 ± 0.023 mm, σ 0.0076667 mm
= 63.023 mm

bore D_s

[45] v_{Dg} = mid ± half-band, σ (normal)
= 57.5 mm nominal → mid 57.463 ± 0.037 mm, σ 0.012333 mm
= 57.463 mm

piston groove root D_g

[46] v_{Dc} = mid ± half-band, σ (normal)
= 63 mm nominal → mid 62.955 ± 0.015 mm, σ 0.005 mm
= 62.955 mm

piston outside diameter D_c

[47] v_b = mid ± half-band, σ (normal)
= 4.7 mm nominal → mid 4.75 ± 0.05 mm, σ 0.016667 mm
= 4.75 mm

groove width b

[48] $f_c = (1 + \alpha \cdot (T_{min} - 20\ °C)) \cdot (1 + s_{min})^{(1/3)}$
$= (1 + 0.00015 \cdot (-25 - 20)) \cdot (1 + 0)^{(1/3)}$
= 0.99325

Ring size factor, cold end

[49] $g_c = 1 + \alpha_m \cdot (T_{min} - 20\ °C)$
$= 1 + 0.000012 \cdot (-25 - 20)$
= 0.99946

Groove metal size factor, cold end

[50] $f_h = (1 + \alpha \cdot (T_{max} - 20\ °C)) \cdot (1 + s_{max})^{(1/3)}$
$= (1 + 0.00015 \cdot (70 - 20)) \cdot (1 + 0.04)^{(1/3)}$
= 1.0208

Ring size factor, hot end

[51] $g_h = 1 + \alpha_m \cdot (T_{max} - 20\ °C)$
$= 1 + 0.000012 \cdot (70 - 20)$
= 1.0006

Groove metal size factor, hot end

[52] $stretch_{nom}$ = $D_{ref}/d - 1$ at the mean dimensions
$= 57.463/56 - 1$
= 2.6125 %

Installed stretch of the ring ID on the seat, mean dimensions

[53] $stretch = D_{ref,max}/d_{min} - 1$
$= 57.5/55.55 - 1$
= 3.5104 %

Largest installed stretch

[54] $stretch_{min} = D_{ref,min}/d_{max} - 1$
$= 57.426/56.45 - 1$
= 1.729 %

Smallest installed stretch

[55] red_{nom} = r(X) at the mean installed stretch
= r(2.6125 %)
= 1.2812 %

Cross-section reduction at the mean installed stretch

[56] $u_{stretch}$ = stretch / x_{hi}
= 3.5104 % / 6 %
= 0.58506

Largest installed stretch against its limit

[57] $X_c = g_c \cdot D_{ref,max} / (f_c \cdot d_{min}) - 1$
$= 0.99946 \cdot 57.5 / (0.99325 \cdot 55.55) - 1$
$= 4.1575\ \%$
Largest stretch at the cold end
[58] $r_c = r(X_c)$
$= r(4.1575\ \%)$
$= 2.0161\ \%$
Cross-section reduction at the cold end
[59] $Wc_{lo} = f_c \cdot W_{min}$
$= 0.99325 \cdot 3.4$
$= 3.377\ mm$
Smallest ring section at the cold end
[60] $Weff_{lo} = Wc_{lo} \cdot (1 - r_c)$
$= 3.377 \cdot (1 - 0.020161)$
$= 3.309\ mm$
Smallest effective ring section
[61] $h_{hi,c} = g_c \cdot (\text{depth stack})\ \text{maximum}$
$= \text{mean } 2.7785\ mm, \text{worst-case maximum}$
$= 2.8085\ mm$
Largest gland depth at the cold end
[62] $squeeze_{min} = 1 - h_{hi,c} / Weff_{lo}$
$= 1 - 2.8085 / 3.309$
$= 15.125\ \%$
Smallest squeeze
[63] $A_{g,hi} = b_{max} \cdot h_{max} - 2(1 - \pi/4) \cdot (g_c \cdot r_g)^2$
$= 4.7974 \cdot 2.8085 - 0.026796$
$= 13.447\ mm^2$
Largest groove area at the cold end
[64] $fill_{min} = (\pi/4) \cdot Weff_{lo}^2 / A_{g,hi}$
$= (\pi/4) \cdot 3.309^2 / 13.447$
$= 63.953\ \%$
Smallest gland fill
[65] $X_h = g_h \cdot D_{ref,min} / (f_h \cdot d_{max}) - 1$
$= 1.0006 \cdot 57.426 / (1.0208 \cdot 56.45) - 1$
$= -0.28\ \%$
Smallest stretch at the hot end
[66] $r_h = r(X_h)$
$= r(-0.28\ \%)$
$= -0.14029\ \%$
Cross-section reduction at the hot end
[67] $Wh_{hi} = f_h \cdot W_{max}$
$= 1.0208 \cdot 3.6$
$= 3.6747\ mm$
Largest ring section at the hot end
[68] $Weff_{hi} = Wh_{hi} \cdot (1 - r_h)$
$= 3.6747 \cdot (1 - -0.0014029)$
$= 3.6799\ mm$
Largest effective ring section
[69] $h_{lo,h} = g_h \cdot (\text{depth stack})\ \text{minimum}$
$= \text{mean } 2.7817\ mm, \text{worst-case minimum}$
$= 2.7516\ mm$
Smallest gland depth at the hot end

[70] $squeeze_{max}$ = 1 − $h_{lo,h}/Weff_{hi}$
= 1 − 2.7516/3.6799
= 25.225 %
Largest squeeze
[71] $A_{g,lo}$ = $b_{min}\cdot h_{min}$ − $2(1-\pi/4)\cdot(g_h\cdot r_g)^2$
= 4.7028·2.7516 − 0.026857
= 12.914 mm²
Smallest groove area at the hot end
[72] $fill_{max}$ = $(\pi/4)\cdot Wh_{hi}^2\cdot(f_h/g_{ref,h})/A_{g,lo}$
= $(\pi/4)\cdot 3.6747^2\cdot(1.0208/1.0006)/12.914$
= 83.783 %
Largest gland fill (unstretched section, volume basis: the held ring takes its whole volume change in its section)
[73] $squeeze_{nom}$ = 1 − h/W_{eff} at the mean dimensions
= 1 − 2.78/3.4552
= 19.541 %
Squeeze at the mean dimensions, installed
[74] $sq_{min,exact}$ = exact non-linear corner (check)
= corner chosen by the sensitivities
= 15.18 %
Check: smallest squeeze at the exact corner
[75] $sq_{max,exact}$ = exact non-linear corner (check)
= corner chosen by the sensitivities
= 25.176 %
Check: largest squeeze at the exact corner
[76] gap_{max} = largest gap over the installed, cold and hot temperatures (diametral clearance, each part with its own expansion) plus the bore dilation under pressure
= mean 0.068 mm, worst-case maximum
= 0.10606 mm
Largest extrusion gap, diametral for a radial seal and axial for a face seal
[77] gap_{min} = smallest gap over the temperatures
= mean 0.068 mm, worst-case minimum
= 0.029984 mm
Smallest extrusion gap
[79] gap_{allow} = permitted extrusion gap
= entered 0.2 mm
= 0.2 mm
Permitted extrusion gap, diametral for a radial seal and axial for a face seal (your record or value)
[80] u_{gap} = gap_{max} / gap_{allow}
= 0.10606 / 0.2
= 0.53032
Largest gap against the permitted gap
[81] $p_{sq,min}$ = fraction of 20000 samples with squeeze below the lower limit (cold end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: squeeze below the lower limit (cold end)
[82] $cpk_{sq,min}$ = limit distance over three σ of the linearised model (the Monte Carlo itself runs the exact model)
= σ = 0.0080809
= 1.5612
Capability index C_{pk}: squeeze below the lower limit (cold end)

[83] $p_{sq,max}$ = fraction of 20000 samples with squeeze above the upper limit (hot end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: squeeze above the upper limit (hot end)
[84] $cpk_{sq,max}$ = limit distance over three σ of the linearised model (the Monte Carlo itself runs the exact model)
= σ = 0.0077697
= 3.4698
Capability index C_{pk}: squeeze above the upper limit (hot end)
[85] $p_{fill,max}$ = fraction of 20000 samples with gland fill above the upper limit (hot end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: gland fill above the upper limit (hot end)
[86] $cpk_{fill,max}$ = limit distance over three σ of the linearised model (the Monte Carlo itself runs the exact model)
= σ = 0.01515
= 1.648
Capability index C_{pk}: gland fill above the upper limit (hot end)
[87] $p_{fill,min}$ = fraction of 20000 samples with gland fill below the lower limit (cold end)
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: gland fill below the lower limit (cold end)
[88] $cpk_{fill,min}$ = limit distance over three σ of the linearised model (the Monte Carlo itself runs the exact model)
= σ = 0.01376
= 2.3815
Capability index C_{pk}: gland fill below the lower limit (cold end)
[89] $p_{stretch}$ = fraction of 20000 samples with installed stretch outside its limits
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: installed stretch outside its limits
[90] $cpk_{stretch}$ = limit distance over three σ of the linearised model (the Monte Carlo itself runs the exact model)
= σ = 0.0027575
= 3.158
Capability index C_{pk}: installed stretch outside its limits
[91] p_{gap} = fraction of 20000 samples with extrusion gap above the permitted gap
= Monte Carlo of the exact model, seed 42
= 0 %
Monte Carlo probability: extrusion gap above the permitted gap
[92] cpk_{gap} = limit distance over three σ of the linearised model (the Monte Carlo itself runs the exact model)
= σ = 0.0091585
= 4.8028
Capability index C_{pk}: extrusion gap above the permitted gap
[93] pareto = largest contributor to the squeeze spread (worst case)
= ring cross-section W 66.373 %; ring inside diameter d 9.3264 %; bore D_s 9.6019 %; piston groove root D_g 14.699 %
= "ring cross-section W"
Contribution Pareto

5. Results

SYMBOL	VALUE	DESCRIPTION
$squeeze_{min}$	15.13 %	Smallest squeeze
$squeeze_{max}$	25.22 %	Largest squeeze
$fill_{max}$	83.78 %	Largest gland fill
stretch	3.510 %	Largest installed stretch
$squeeze_{nom}$	19.54 %	Squeeze at the mean dimensions
$fill_{min}$	63.95 %	Smallest gland fill
$stretch_{nom}$	2.612 %	Installed stretch, mean dimensions
$stretch_{min}$	1.729 %	Smallest installed stretch
X_c	4.158 %	Largest stretch at the cold end
red_{nom}	1.281 %	Cross-section reduction at the mean stretch
gap_{min}	0.02998 mm	Smallest extrusion gap
gap_{max}	0.1061 mm	Largest extrusion gap, diametral for a radial seal (over the temperature range, with the bore dilation)
gap_{allow}	0.2000 mm	Permitted extrusion gap, diametral for a radial seal
$p_{sq,min}$	0 %	Fraction outside the limit: squeeze below the lower limit (cold end)
$cpk_{sq,min}$	1.5612	C_{pk}: squeeze below the lower limit (cold end)
$p_{sq,max}$	0 %	Fraction outside the limit: squeeze above the upper limit (hot end)
$cpk_{sq,max}$	3.4698	C_{pk}: squeeze above the upper limit (hot end)
$p_{fill,max}$	0 %	Fraction outside the limit: gland fill above the upper limit (hot end)
$cpk_{fill,max}$	1.648	C_{pk}: gland fill above the upper limit (hot end)
$p_{fill,min}$	0 %	Fraction outside the limit: gland fill below the lower limit (cold end)
$cpk_{fill,min}$	2.3815	C_{pk}: gland fill below the lower limit (cold end)
$p_{stretch}$	0 %	Fraction outside the limit: installed stretch outside its limits
$cpk_{stretch}$	3.158	C_{pk}: installed stretch outside its limits
p_{gap}	0 %	Fraction outside the limit: extrusion gap above the permitted gap
cpk_{gap}	4.8028	C_{pk}: extrusion gap above the permitted gap
$u_{stretch}$	0.59	Largest stretch against its limit
u_{gap}	0.53	Largest gap against the permitted gap

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Squeeze minimum against the lower band limit	$squeeze_{min}$ 15.125 %	≥ 15 %	–	[5](record)
PASS	Squeeze maximum against the upper band limit	$squeeze_{max}$ 25.225 %	≤ 30 %	–	[5](record)
PASS	Gland fill maximum against the upper band limit	$fill_{max}$ 83.783 %	≤ 85 %	–	[5](record)
PASS	Gland fill minimum against the lower band limit	$fill_{min}$ 63.953 %	≥ 60 %	–	[5](record)
PASS	Installed stretch against the upper limit	stretch 3.5104 %	≤ 6 %	–	[6](rule)
PASS	Installed stretch at the cold end against the upper limit	X_c 4.1575 %	≤ 6 %	–	[5](record)
PASS	Installed stretch against the lower limit	$stretch_{min}$ 1.729 %	≥ 0 %	–	[5](record)
PASS	Extrusion gap against the permitted gap	gap_{max} 0.10606 mm	≤ 0.2 mm (entered 0.2 mm)	–	[7](record)

7. Applicability

WARNING records: tabulated values not taken from a reference record are α, ν (Material properties); d_r, $d_{r,tol}$, W, W_{tol} (Ring size); $q_{lo,s}$, $q_{hi,s}$, $q_{lo,d}$, $q_{hi,d}$, $fill_{lo}$, $fill_{hi}$, x_{lo}, x_{hi}, od_{hi} (Squeeze, fill and stretch criteria); gap_{allow} (Permitted extrusion gap). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

```
Template fp.oring-gland@1.5.0 Content hash 9178aeb47bc3978b Engine calc-engine/1.0.0
Known-answer cases passed 13/13 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Parker O-Ring Handbook ORD 5700.
2. Parker PTD 5705.
3. ISO 3601-1:2012.
4. ISO 3601-2:2016.
5. oring.criteria.
6. oring.criteria record; PTD 5705 installation.
7. oring.extrusion.